Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-region Snowflake warehouse scaling	Potential query latency under concurrent peak loads	High
2	Static batch ingestion intervals	Delay in processing real-time practice session updates	Medium
3	Local Streamlit state caching limits	Memory overhead during heavy multi-tab visualization rendering	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single instance Streamlit deployment	Containerized Kubernetes auto-scaling across multi-region cluster
2	Data Storage	Standard Snowflake multi-cluster warehouse	Automated data partitioning and cold storage tiering for historical s
3	Performance	Standard unindexed query execution	Implementation of materialized views and Redis caching layer
4	Security	Role-based access control (RBAC)	Integration with Enterprise SSO, OAuth 2.0, and column-level enc

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Real-time live race telemetry WebSocket stream integration	Q3 2026	Sub-second live insights

Phase 2			
Phase 3	AI-driven predictive pit stop strategy and tire wear forecasting model	Q4 2026	Enhanced tactical decision-making
Phase 4			

Phase 4 Mobile companion app with push notifications for race alerts Q1 2027 Broader user engagement